

SUMON SOURABH GHOSH

SDET – Backend & DevOps Focus

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Backend-focused SDET with 2 years of experience at TCS specializing in API testing, backend validation, and CI/CD pipeline automation using pytest and requests. Proficient in Python and Node.js; hands-on with Linux, AWS (EC2, S3, CloudWatch), Docker, Terraform (IaC), and automated deployment workflows. Experienced in TDD/BDD methodologies, security-aware API validation, and performance testing (k6). Proven ability to validate complex distributed systems, debug production issues, and drive reliability through automation, Blue/Green deployment strategies, and continuous delivery.

WORK EXPERIENCE

Tata Consultancy Services — Assistant System Engineer

Jun 2022 – Feb 2024

- Developed and maintained backend services using Node.js and REST APIs; implemented comprehensive API validation, request/response analysis, and end-to-end workflow testing using the requests library, incorporating security checks for common vulnerabilities (auth bypass, injection, improper error handling).
- Designed and authored 300+ test cases (functional, regression, exploratory, and BDD-style using Gherkin) with pytest across distributed microservices, ensuring contract adherence and inter-service communication reliability.
- Configured and maintained CI/CD pipelines using GitHub Actions and Jenkins, automating build, test, and deployment stages with Blue/Green deployment strategies, reducing deployment lead time by 30%.
- Applied data-consistency validation scripts across distributed systems, identifying 30+ edge-case failures and reducing data-integrity defects by 25%.
- Provisioned and managed AWS infrastructure (EC2, S3, IAM) using Terraform (IaC), reducing manual environment setup time by 40% and ensuring cross-environment consistency.
- Spearheaded resolution of production incidents using CloudWatch logs and root-cause analysis, cutting mean time-to-resolution by 20% and reducing critical post-deployment bugs by 15%.
- Contributed to backlog refinement, story estimation, and sprint retrospectives within Agile/Scrum; mentored 3 junior engineers on test automation best practices and CI/CD workflows.

PROJECTS

applier.cloud – AI Resume Builder SaaS

2025 – Present | applier-cloud.netlify.app | github.com/sghoshm/applier.cloud

- Architected and deployed a full-stack AI Resume Builder SaaS using the MERN stack (MongoDB Atlas, Express, React, Node.js) with Groq LLM API (Llama 3.3 70B) for resume tailoring and cover letter generation.
- Implemented JWT + Google OAuth 2.0 authentication with bcrypt password hashing, refresh token rotation, and Passport.js; secured API routes with role-based plan middleware (Free/Pro/Elite).
- Built PDF parsing pipeline using pdf2json with AI-assisted structured extraction, and PDF export using @react-pdf/renderer with dynamic resume templates.
- Designed ATS scoring system that analyzes resume–job description keyword alignment and returns match score, missing keywords, and improvement suggestions via LLM.
- Deployed backend on Render and frontend on Netlify with CI/CD via GitHub; configured MongoDB Atlas M0 free cluster — full infrastructure at \$0/month.
- Designed freemium SaaS pricing model (Free/Pro/Elite) with AI credit system, plan-gating middleware, and Lemon Squeezy payment integration scaffold.

OpenEarth – Data Aggregation & Visualization Platform

2025 – Present | earthint.netlify.app | github.com/sghoshm/open-earth

- Developed a full-stack platform aggregating Earth-related datasets via multiple external APIs; validated all API responses for accuracy, consistency, and security using automated pytest scripts.
- Engineered backend data-retrieval and processing logic; implemented interactive frontend for visualization.
- Deployed via automated CI/CD pipelines on Netlify with health monitoring; evaluated system behaviour under 15+ diverse input and edge-case scenarios.

Nexus OSINT Platform

2025 – Present | nexusint.netlify.app | github.com/sghoshm/nexus-osint

- Constructed backend system aggregating real-time public data from multiple APIs; containerized with Docker and managed deployments via GitHub Actions CI/CD pipelines.
- Designed modular architecture following microservices principles; optimized data-pipeline performance and applied structured validation across all data sources.

OpenJarvis – AI Assistant with API Integration & Automation

2025 – Present | github.com/sghoshm/open-jarvis

- Built a Python AI assistant integrating external APIs, OS-level automation, and dynamic command workflows with safety checks.
- Validated API responses for accuracy and error handling; completed end-to-end testing across 20+ input scenarios following TDD principles, covering command-parsing, execution, and response flows.

Personal Project: Self-Hosted Video Streaming Platform

2020 – 2024

- Designed and deployed full-lifecycle backend (auth, session management, media delivery) on Linux with SSL and Cloudflare tunnelling; managed DNS routing and secure traffic configuration.
- Owned full development, deployment, testing, and monitoring pipeline; performed load and scalability analysis under concurrent traffic using k6.

Hand Gesture Controlled Mouse (Python)

2023 | github.com/sghoshm/hand-gesture-mouse

- Developed real-time computer-vision gesture-recognition system; optimized frame-processing performance and validated edge-case inputs systematically.

EDUCATION

Bengal Institute of Technology and Management

2024 – 2026

M.E / M.Tech – Electronics & Communication Engineering

Sathyabama Institute of Science and Technology

Aug 2018 – Sep 2022

B.E / B.Tech – Electronics & Communication Engineering

TECH STACK & SKILLS

Testing & QA: API Testing (Postman), Functional / Regression / Exploratory Testing, Test Case Design, BDD / TDD Principles, Gherkin-style Test Design, Backend Validation, Security-Aware API Validation, Performance Testing (k6), Data & Edge-Case Testing, Defect Lifecycle Management, pytest, requests, API automation scripting

Programming: Python, JavaScript (Node.js)

Backend & APIs: REST APIs, JWT Authentication, API Validation, Microservices (contract & inter-service testing), Node.js

DevOps & Infra: Linux Server Administration, AWS (EC2, S3, CloudWatch, IAM, Lambda, RDS), Docker, CI/CD (GitHub Actions, Jenkins), Blue/Green & Canary Deployment Strategies, Infrastructure as Code (Terraform), SSL/HTTPS, Cloudflare DNS & Tunnelling

Databases: MongoDB, Firebase, PostgreSQL / SQL (queries, data extraction & validation)

Tools: Git, GitHub, Postman, k6, CloudWatch